

$I \in \mathbb{R}^{3\times 352^2}$

448^2

DINOv3
ViT-S/16 ✳
+ Conv-LoRA
in all 12 blocks
(rank 16, trained)

384×28^2

Multi-depth
readout
1×1 conv
to 192 ch.

T_3, T_6, T_9, T_{12}

Token pyramid
one depth
per stride

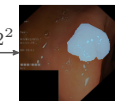
s_4–s_{32}

Scale-aware
decoder
residual DW
+ top-down

deep supervision at s_8–s_{32}

Z_{SAD}

$\uparrow 352^2$



$\hat{Y} = \sigma(Z)$

T_6, T_9, T_{12}

Dual-prototype
margin head
prototypes,
anchors,
self-support

$0.2\,\pi_{\mathrm{proto}}$, upsampled

π_{proto} on the 28×28 grid

proposed ✳ frozen - - → training only